

SV-03 NodeOS Layered Architecture

Hardware
Independent

Application

Acquisition sequence | System state | Command dispatch

Core Services

Measurement | Calibration | Diagnostics | Communication

Platform Services

TMAG5273 | EEPROM | UART | TWI0 | TWI1 | Watchdog | Timer | GPIO

Hardware Abstraction Layer

Microcontroller-dependent access and register isolation

ATmega328PB

Physical MCU and peripherals

NodeOS Responsibilities

Measurement
Calibration
Diagnostics
Communication
Configuration
Fault recovery

Hardware
Dependent

Design rule: the application layer never directly accesses hardware resources.